



OPERATION Department
SIMHADRI

Note No :1



Ref:SMTTP/OPERATION/2023-24/LMI/715133

Date:

07/12/2023(20:48:56)

Subject:REVISION OF LMI ON FLEXIBLE OPERATION OF COAL FIRED
UNITS

“FLEXIBLE OPERATION OF COALFIRED UNITS” has been revised as per CC OS document-
OIN/OPS/ SYS/ 071/REV.:01NOV'23.

Revised LMI is attached and put up for further perusal and approval

Submitted by:

नवीन सागर - वरिष्ठ प्रबंधक

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On:07/12/2023(20:48:56) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:2

Proposed LMI reviewed and found in line with draft OIN, attached herewith for ready reference. Presentation on the same will be given at DPM on 09.12.2023 by custodian. Hence necessary approval may please be accorded.

Submitted by:

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On:08/12/2023(09:17:21) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:3

Individual sectional heads are requested to review the responsibilities identified for each section as per LMI and comments/changes if any may be recorded before approval

Submitted by:

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On:08/12/2023(10:02:43) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:4

LMI reviewed and submitted for approval

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On:08/12/2023(11:51:29) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:5

May pl review

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On:08/12/2023(15:44:32) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:6

1. For Providing feedback of SADC Actuators in control room Installation of SMART Positioners is required in SADC Actuators. This will improve the Accuracy of SADC Operation and Realtime feedback will be available in Control Room
2. For Flame scanners, Upgradation to Latest Available from BHEL (OEM) BN10 is in Process.
3. For Accurate Sliding Mode of Operation IP Control Valve throttling Issues at low load to be taken care, As large pressure variation in Th. Pr. SP to avoid IP Control Valve throttling leads to disturbances in Process parameters such as Drum level, SH/RH Temperatures and Unit stability in general.
4. Mill Scheduler: Modified SLC of the Mills implemented in Unit #3 during OH. Trial of the same is pending.

Submitted by:

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On:08/12/2023(17:49:41) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:7

Forwarded after LMI review for further approval

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On:08/12/2023(18:12:27) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:8

Review

Submitted by:

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On:09/12/2023(08:57:08) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:9

Please review

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On:12/12/2023(11:44:46) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:10

Reviewed the LMI.

Forwarded for further approval.

Submitted by:

रश्मि रंजन सतपथी - वरिष्ठ प्रबंधक

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On:13/12/2023(09:44:15) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:11

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On:13/12/2023(18:01:26) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:12

Point 13 on Page 12 regarding Nitrogen sparging or Nitrogen blanketing of DM water storage may be omitted. There is no such communication from COS in this regard.

Submitted by:

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On:14/12/2023(16:13:50) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:13

Please review and do necessary changes as recommended by DGM(Chemistry).

Submitted by:

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On:14/12/2023(17:45:00) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:14

Sir, document updated inline with discussion during DPM presentation

Submitted by:

नवीन सागर - वरिष्ठ प्रबंधक

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On:15/12/2023(09:56:44) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:15

For review and further process please.

Submitted by:

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On:15/12/2023(10:03:52) **Ref:**SMTPP/OPERATION/2023-24/LMI/715133

Note No:16

Reviewed & forwarded

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Note No:17

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On:15/12/2023(10:41:07) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:18

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On:15/12/2023(10:49:55) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:19

Please mention the updation/revision details as mentioned in Note no. 14.

Submitted by:

मैथ्यू ई कोवूर - महाप्रबंधक

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On:21/12/2023(17:45:31) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:20

Document updated with revisions and put up for approval

following revisions have been made

1. Updated status of mitigation plan for flexible operation of Simhadri units, in table 17.9 (after DPM discussion & details presented in Note no:6 by C&I)
2. Clause 15.13, DM storage tanks N2 blanketing as good O&M practice, omitted as per Chemistry (Note no: 12)

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On:26/12/2023(12:43:01) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:21

May please be approved.

Submitted by:

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On:27/12/2023(15:21:24) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133

Note No:22

Approved & Submitted by:

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On:27/12/2023(15:29:52) **Ref:**SMTTP/OPERATION/2023-24/LMI/715133



NTPC LIMITED

Simhadri Super Thermal Power Station

LOCATION MANAGEMENT INSTRUCTION

LMI/OIN/OPS/SYST/071

Rev. No.: 2 Dec 2023

Flexible Operation of Coal-fired Units

Approved for
Implementation by

HOP (Simhadri)

Date:

LMI	<i>Flexible Operation of Coal-fired Units</i>		
LMI No	LMI/OIN/OPS/SYST/071		
Issue No	2	Revision: 2	Dt: 07.12.2023

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Checked by	OM PRAKASH	AGM (BMD)	
Checked by	JAYAMOL RACHEL VARUGHESE	AGM (C&I)	
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Rechecked by	MATHEW EIPE KOVOOR	GM (Maint)	
Reviewed by:	DHURJATI PRASAD PATRA	GM (O&M)	
Approved by	SANJAY KUMAR SINHA	HOP (Simhadri)	

RECORD OF REVISIONS

OLD REV NO.	NEW REV NO.	EXISTING CLAUSE	NEW CLAUSE	EXISTING STATEMENT	REVISED STATEMENT
1	2	1.0	1.0	Introduction mentioning India's commitment in COP'21. CEA advisory to identify at least 50% of the capacity for further lowering the MTL below 55%	Introduction statement included as in revised OIN, including mention of CEA mandate of achieving 40% MTL in phased manner
1	2	14.0	14.0	Pilot studies in various power plants	Pilot studies and implementation status updated as in revised OIN
1	2	17.9	17.9	Suggested mitigation plan and action items for flexible operation of the units	Updated status of mitigation plan and action items for flexible operation of Simhadri units
1	2	15	15	15.13 DM storage tanks N2 blanketing as good O&M practice before flexibilization	Point omitted as DM tanks blanketing not present. (No communication to CHEM from COS in this regard)

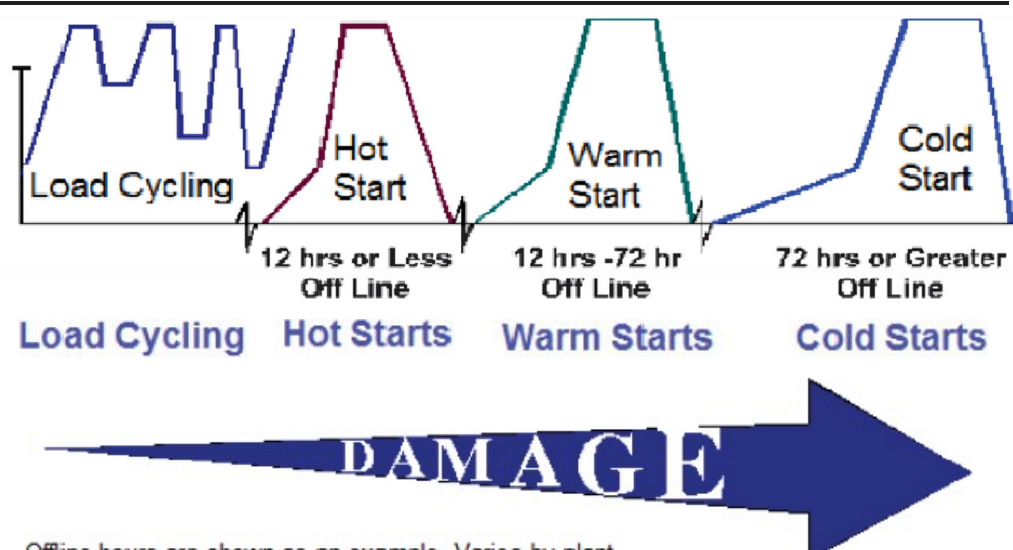
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Flexible Operation of Coal-fired Units

1.0	INTRODUCTION
	In India, coal-based power plants have traditionally supported the major electricity demands of the Indian power grid. Fossil (mainly Indian sourced coal) power plants were running close to base load. Many of these NTPC plants are experiencing already the first indications of a push for more flexible operation; the load factor and utilization rate of these plants is decreasing.
	India has committed to a Nationally Determined Contributions (NDC) target of achieving 40% of its total electricity generation from non-fossil fuel sources by 2030. In line with GoI target of 500 GW, NTPC has also set an ambitious target of 60 GW RE capacity by 2032. The challenge with RE is the high uncertainty and variability in generation and additional resources are required for balancing the variations. Gas and Hydro based stations are designed for fast ramping, but India lacks sufficient quantities of cheap gas and weather dependency of hydro stations render them rather ineffective for meeting flexibilization requirements completely. Battery Energy Storage Systems are in a nascent stage and very expensive. Due to other flexing resources, the coal plants are expected to provide balancing power at the most economical cost.
	Vide gazette notification dated 30 January 2023, CEA has mandated all coal based plants to achieve 40% MTL and ramp rates to be implemented as per CEA phasing plan.
	In NTPC many studies have been carried out in the last few years to identify the constraints in flexible operation up to 40% at higher ramp rates and quantify the investments and costs associated with cycling of the units with reputed consultants.
2.0	REFERENCE DOCUMENT
	Flexible Operation of Coal-Fired Units OIN/OPS/SYST/071, REV.: 01 Nov.2023
3.0	SUPERSEDED DOCUMENTS
	LMI/OIN/OPS/SYST/071Rev. No.: 1

4.0	RECORD OF REVISION
	1. Clause 1.0 Introduction part new statement included as in revised OIN 2. Clause 14.0 Pilot studies and implementation status updated as in revised OIN
5.0	SCOPE:
	Stage-1 & 2 units of Simhadri
6.0	FEATURE OF FLEXIBILISATION
	The main characteristics of flexibilization of coal based thermal power stations are: a) Stable Low/ Minimum Load Operation b) Increased Ramp Rate (Both upward & downward) c) Reduced effect on life of Equipment d) Efficient Operation even at low load e) Minimal impact on environment The intermittency and variability associated with high solar and wind power penetration would increase the need for flexibility in power systems and backup capacity. The flexibility of generating station implies ability: a. To cycle on and off including its lead time (Hot/Warm/Cold Start- up) b. The ramping rate at which it can vary the generation (Increased ramp rate) c. Maximum and Minimum output while it is in operation (Load following and min load operation)

7.0	TYPE OF FLEXIBLE OPERATION The frequent load variation or start-up will stress the equipment's from varying thermal loading and these type of cycling operational. The impact may be categorized into safety hazards, failure of mechanical components, impact on cycle chemistry and finally on efficiency and environment  Offline hours are shown as an example. Varies by plant
8.0	CYCLIC OPERATION-SAFETY CONCERN Sub-optimized combustion and sometimes lean coal air mixture can lead to frequent tripping on flame failure, furnace pressurization, slagging and clinkering. Low flue gas temp below Acid Dew point can lead to duct, APH basket and ESP fields corrosion. Even chances of Collapse of ESP. Due to corrosion boiler drains may also burst leading to unsafe situation.
9.0	CYCLIC OPERATION- IMPACT ON MECHANICAL DAMAGE ○ Fast & Frequent thermal cycling of components lead to fatigue, creep. ○ Stresses on components and turbine shells resulting from changing pressure & temp. Failures of boiler tubes caused by cyclic fatigue, corrosion fatigue and pitting. ○ Cracking in dissimilar metal welds, headers and valves, and other thick-walled components due to rapid changes in steam temperature. ○ Cracking of generator rotors due to the movement between the rotor and casing during "barring".

10	CYCLING AND FAST RAMPING- CYCLE CHEMISTRY
	Corrosion caused by oxygen entering the system (e.g., during start-up), and changes to water quality and chemistry, resulting from, e.g. falling pH. Condensation from cooling steam, which in turn can cause corrosion of parts, leakage of water, and an increased need for drainage. Oxidation, e.g, from exposure to air on start-up and draining; oxides in boiler tubes can dislodge due to thermal changes. Corrosion of turbine parts, not only from oxides but also from wet steam that occurs on start-up, during low-load operations.
11	CYCLING AND FAST RAMPING- IMPACT ON HEAT RATE AND ENVIRONMENT
	○ Heat rates typically degrade at partial load due to lower steam parameter (Pressure and temp) and Low Feed water temp at ECO inlet. ○ NOX and SO2 rates are also affected by loading. Startup emissions of CO2, NOX, and SO2 may be significantly higher than steady-state emissions rates. ○ Ramp ups in power output may also result in higher than steady-state emissions. Low flue gas temp below acid Dew point can damage APH basket and flue gas duct.
12	IMPACT ON BOILER TUBE FAILURE
12.1	Tube to header connections of all modules will be more frequently stressed by possible restrained thermal expansion. Theoretically, the construction of the assemblies is such as the tubes can freely expand and/or have sufficient flexibility to avoid stress concentration at the header connection. Nevertheless, even in base-load mode, tube-header connection failures are typical damages (not always only because thermo-mechanical stresses, weld quality plays also an important role). In the cycling scenario, these connections will be

	more often stressed and low cycle fatigue is expected to occur. Visual inspection of proper alignment of tubes/harps can help identify zones/tubes at risk. DPT/MPI is then conducted to find possible crack initiation. When welding new connections (for repair or replacement), care must be taken to the quality of the weld (avoid notch or weld defect).
12.2	Fire side corrosion- could slightly increase if the temperature in the boiler allows condensation of water or acid during shut-down periods. The zones at risk are the coldest locations. Visual inspections of the boiler internal condition can identify critical locations. Later, temperature measurements could confirm the cold spots.
12.3	The major concern in the high temperature tubes is the thickness of the internal oxide layer observed on several unit. When going to a cycling operating mode, because of the differential thermal expansions' coefficient between the oxide and the metal and because of more frequent temperature transients, thick oxide layers will peel-off/exfoliate/break-off. Solid hard oxide particles will flow with the steam and cause solid particle erosion (SPE) on downstream components (tubes, main steam valves internals, strainers, drain valves, ST blades and vanes). Oxide deposits can also accumulate and induce subsequent damages (plugged tubes will be poorly cooled down and will fail by short term overheating; fouled dead zones as sampling and drains lines will cause malfunction of these systems; probes can be covered with oxide and become unrepresentative of the process parameters).
12.4	Dissimilar weld joint failure - The large difference in coefficient of thermal expansion between ferritic and austenitic alloys causes a significant stress concentration along the interface, which can result in fatigue failures at these welds. Increasing the number of start/stops will therefore have a negative impact on these weld connections.

13	IMPACT ON STEAM TURBINE
	○ The actual impact of the weekly cycling scenario on the ST components will, for a large part, be the same as for the low load scenario. ○ Water droplet erosion on the last stages blades and LP casing will occur. If the average load is higher, the erosion rate will be lower. It could have a positive impact compared to the low load scenario. ○ Wear and solid particle erosion on the first stages of the HP and IP turbines, strainers and valves internals will increase if the magnetite issue in the RH and SH is not solved. Wear and tear of valves will increase. ○ Thermo-mechanical damage (thermal shocks, fatigue) should not be an issue if the turbine is operated according to the OEM recommendations. In case of excessive thermal stress, cracks could occur in massive components (casing, valves bodies, rotors). ○ More frequent run-ups and run downs mean that the steam turbine will pass the critical speed for each speed transient. Once again, this is, in theory, not a problem provided that the vibration behavior is acceptable, stable and properly monitored. If high vibration occurs, the first damages are expected on the bearings and in steam glands (small clearances). When corrective action is taken in due time like balancing, alignment, maintenance of lube oil system, minor damage repair, seal replacement etc. ○ To mitigate and enable units for flexible operation the stations O&M practices are very much important apart from investment in capital addition required for minimum load operation and frequent start up and higher ramp rate for flexing.

14.0	PILOT STUDIES FOR LOWER 'MINIMUM TECHNICAL LIMIT' AND HIGHER RAMP RATE			
Associates	Plant	Study period	Scope	
IGEF/VGB, Germany	Dadri Unit 6	Jan- Aug 2018	MTL (40%) & 3% Ramping	
GE, USA	Talcher-K Unit 1	April-Aug 2018	MTL (40%)	
USAID/BHEL	Mauda Unit 2	April-Aug 2019	MTL (40%)	
J-Coal/TEPCO, Japan	Vindhyachal Unit 11, Mauda Unit 3,4	March-July 2019	MTL 55% with 3% ramping. Efficiency at different part load condition	
EPRI, USA	Solapur, Gadarwara, Simhadri, Bongaigaon, Farakka and Unchahar	Mar-Dec'20	Identifying gap in intended O&M practices to be followed in view of flexible operation of fleet	
USAID/INTERTEK, USA	Ramagundam Unit 2 & Jhajjar Unit 1	Jan-July 2018	Cost of flexing due to start up and load following	
Engie Lab France/Belgium	Dadri Unit 4 & Farakka U6	Nov 18-Sept 19	Estimate Capital Cost to enable flexibility & increase in running cost due to load ramping and startups.	
<i>In all the Pilot Studies, field tests were conducted in strictly controlled environment with good coal quality in presence of functional experts. The following technical issues were observed.</i> ▪ Excessive fluctuations in Steam temperatures ▪ High Drum level swings during ramping ▪ Flame disturbance during ramping and at MTL ▪ Occasional furnace pressurization ▪ Stalling of Primary Air fans at low loads ▪ Low boiler flue gas exit temperature, leading to acid corrosion. ▪ With 3% ramp commands, actual ramp rate achieved in full load range was only 1.3% due to manual milling system operation. Two studies were conducted with M/S Intertek, USA (Ramagundam U2 & Jhajjar U1) and M/S Engie Lab (Dadri U4 & Farakka U6) to quantify the costs associated with cycling of the units. Various factors influencing the costs include the automation Levels in the plant, coal quality, age & size of Units, remaining component life, design of the machine, maintenance philosophy, preservation practices, water chemistry controls, operational expertise and extent of cyclic operation. Contract was awarded to M/S Siemens for Dadri Unit 6 and M/S Siemens & M/S Emerson for Simhadri Units for implementation of measures for Minimum loading operation and higher ramp rates. Various software solutions and hardware retrofits have been implemented.				

15.0	GOOD O&M PRACTICES BEFORE INTRODUCING FLEXIBILIZATION IN COAL BASED UNITS	
	It is essential that some basic practices are followed before preparing a unit for flexing. The following list is indicative only and contains the broad preparation items.	
sl.no	O&M PRACTICES	Responsibility
1	All auto loops should be available and fine tuning of CMC must be carried out to minimize the deviation of parameters like MS/HRH steam temperature, throttle pressure, drum level, excess O2% at economizer outlet and flue gas temperature at boiler outlet.	C&I, OPN
2	Attemperator system (isolating valves and control valves) and control valves are to be set tight and must give fast response to the changing system demand.	C&I, BMD, OPN
3	Optimize minimum coal loading in a mill by fine tuning primary air flow vs coal flow curve to avoid lean air mixture and possible flame failure tripping.	OPN, EEMG, BMD, C&I
4	Dirty air flow test at regular interval to evaluate partially plugged coal pipes and burners	EEMG
5	Burner tilts should be operational in full range in auto mode and SADC damper operation should be checked and correct feedback must be made available.	C&I, BMD, OPN
6	WWSB and LRSB operation scheduling should be done at higher load during such opportunity.	OPN
7	Air heater soot blowing must be carried out at least once in a shift	OPN
8	Air heater air leakages and other tramp air should be minimized.	BMD, EEMG
9	Replacement or repairing of expansion joints if required and major duct revamping if any.	BMD
10	Water chemistry instrumentation should be set right and linked with DCS.	CHEMISTRY, C&I
11	SCAPH operation to be made through to contain flue gas temperature less than acid dew points	BMD, OPN C&I
12	Check condenser- air- leak. Helium leak detection may be deployed	EEMG, OPN, TMD
13	Boiler side high energy piping hanger indicator are to be marked and monitored.	BMD
14	Low load FRS to be used to reduce flow rate in economizer during cold or warm start up.	OPN
15	Turbine stop and control valves to be inspected.	TMD
16	Ensuring availability of deaerator pegging heating with auxiliary steam sources.	TMD, OPN

17	Flame scanners to be maintained in perfect working condition all the time	C&I
16	PRACTICES TO BE FOLLOWED TO MINIMISE EFFICIENCY LOSS	
sl.no	DESCRIPTION	Responsibility
1	Online TLA is to be installed in the Unit for easy access to operating engineers.	EEMG
2	Steam pressure to be maintained as per modified sliding pr. curve (5-10 KSC higher than sliding pr.)	OPN, C&I
3	MS and HRH steam temp to be maintained as per rated by fine tuning of attemperator control	OPN, C&I
4	Keep condenser online tube cleaning system operational. Condenser DO tend to rise at low loads	TMD, OPN, EEMG
5	Keep SCAPH in service to maintain flue gas temp. above acid dew points. All APH dampers to be made operational and reliable.	OPN, BMD
17	SUGGESTED SOLUTIONS	
	The following solutions can help address the issue of parameter deviations like steam temp, drum level, furnace pressure etc. during the Pilot Studies on flexible operation of the units.	
17.1	Fine tuning of existing CMC logics	
	The responses of the existing CMC logics resulted in high deviations in parameters during the tests conducted at 3%/minute ramp rate. These deviations in actual ramp rate achieved can be mitigated by fine tuning of existing control loops. CMC logic loops need to be tuned for controls at low load condition of 40% TMCR. Also, the load ramping for entire new operating range from 40% TMCR to 100% TMCR also needs to be established with CMC Ramp rate of 3% without much deviation in Steam temp parameters	
17.2	Steam Temperature Control:	
	Apart from existing control loop tuning, Advance Process Control can assist in reducing parametric variations, especially Steam Temperatures. During the tests, the main steam temperature deviations were very high, and this can potentially result in higher lifetime consumption of boiler parts. The major task of the temperature controller is to achieve stable steam temperatures using a load dependent set point in all load situations. In normal load operation, set point changes occur very rarely. Disturbances have to be compensated for quickly, so as to allow the unit to be operated close to the material limits in the most critical situation.	

	From a control point of view, the reheat steam temperature is difficult to handle due to: • its sluggish and extremely non-linear dynamic behavior • the nonlinearities of the water/steam table • the non-linear behavior of the attemperator injection valves • non-measurable disturbances from the flue gas The reheat steam temperature should be controlled using burner tilts as part of the automated control, else if the tilts are operated manually, reheat steam temperatures drop during low load operation. During the Studies, further test runs were conducted to investigate the influence of the burner tilts as well as the design and integration of the logic for automated reheat steam temperature control.
17.3	Unit Control and Automatic Mill Operation (Mill Scheduler)
	The main task of the unit control is to provide set points for the steam generator and the turbine, which meet specific requirements defined by the operator or load dispatcher. The two main variables - steam pressure and unit load - have to be controlled by the slow acting boiler and the fast-acting turbine. The dynamic behavior of the plant is replicated using a simplified model of the unit dynamics, which only includes the components for boiler dynamics and steam storage. An additional task of the unit control is to take fans, boiler feed pumps and mills automatically into operation during load ramps in order to allow smooth and uninterrupted load changes. A mill scheduler is subordinate to the unit control and switches coal mills ON / OFF automatically depending on the firing demand and the actual number of firing devices in service. The center line for the firing devices in service can be specified. This enables the boiler's firing balance point to be ascertained, e.g. depending on the start-up conditions of the boiler (cold, warm or hot start). The system also has an automatic replacement strategy should one mill not go into service or fail during service.
17.4	Flue Gas Temperature Control
	The steam air preheater (SCAPH) should be taken into operation automatically, when needed. This control combined with the upgraded temperature control would prevent corrosion in rotary air preheaters.
17.5	Boiler Fatigue Monitoring System:
	The Boiler Fatigue Monitoring System can determine the residual lifetime of highly stressed components by calculating the creep and low-cycle fatigue of specific components during real-time operation. These components are water and steam piping systems with limited service- life, such as boiler headers, drums, separator, attemperators, and piping. The system enables deviations to be detected online and early on, based on real time signals and active management of an operating database. The benefits include the following.

	• Transparency in operating mode on residual life • Detection of high-wear operating modes • In-time notifications for overhaul and inspection • Enhanced power plant safety and reliability • Utilization of component material reserves • Cost-effective in-service monitoring and analysis			
17.6	EOH (Equivalent Operating Hours):			
	The EOH (Equivalent Operating Hours) concept provides an overview of the life consumption of standard operating hours as well as of equivalent start up hours (ESH) that reflect load changes and actual stress on the turbine components being subjected to ramp up and ramp down. The so-called equivalent start-up hours ESH are calculated from temperature differences in thick-walled turbine components arising during turbine start-ups, shutdowns and load changes with distinct steam temperature changes. Hence the ESH represent the turbine service life expenditure caused by temperature induced stresses.			
17.7	Feed water recirculation valve:			
	For stable operation of TDBFPs, the feed water recirculation valves need to be replaced from ON/OFF valves to modulating type control valves. The new feed water control valves would be kept full open for base load operation.			
17.8	Combustion optimization using an online coal flow measurement & balancing system			
	An online coal flow measurement system provides detailed information about the coal distribution between mills and the coal pipes, and a real time balancing system enables combustion to be optimized at all load conditions.			
17.9 The following table summarizes the possible challenges, mitigation plan and action items for flexible operation of the units				
S No	Challenges	Mitigation plan	Actions Required	Status in Simhadri
1	Achieving overall Ramp rate>1% from tech min to full load frequently during the day without any major parameter deviation.	Addressing excursions of boiler metal temp and MS/HRH steam temperatures swing and Pressure swing, drum level fluctuation.	Control loop tuning, modified sliding mode of operation, Milling system automation, Replacement of BFP R/C valves with regulating type.	1% ramp being achieved. Automation of Milling system in U-3 & Replacement of BFP R/C valves with regulating type done in all units.

2	Minimum loading operation- Difficult to reduce load below 55% MCR without oil support	Establish program; Implement a Systematic Approach to Minimum Load Reduction	Control loop tuning up to 40% MTL with modified sliding pressure mode milling automation,	Modified sliding pressure mode of operation available.
3	Flow accelerated corrosion in Economiser tube and Feed water temp reaching saturation temp corresponding to drum pressure	Introduction of program of Flow Accelerated corrosion (FAC). Modified Sliding pressure mode operation.	VFD in LP Dosing pump to maintain PH.	Stroke adjusted Dosing pumps available
4	Steam Temperature control needs improvement following synchronization to mitigate reported excursions in major SH/RH components; Mismatch in heat pick up in MS left/right	Close monitoring of deviation of MS/HRH and metal temp during ramping.	Advance process control of steam/water cycle and load control, SH/RH spray control valves upgrade	Stage-2 completed. Stage-1 implementation is in progress.
5	APH gas temperature below acid dew point.	SCAPH to be made through during low load operation.	Automation in SCAPH control to maintain APH gas temp	Control loop available
6	Drum level swing during ramping and low load.	Auto loop tuning of feed water cycle	Single pump operation and installing regulating type recirculation valves	FRS low range CV available
7	PA fan Stalling	Continuous tracking of PA fan characteristic curve and provisioning of alarm before fan operation near the stalling zone	Single PA fan operation in automated mode at low load.	PA fan stall curve Available. Single PA fan Operation in automated Mode at low load to be Done.
8	Thermal Fatigue and Creep Damage in thick-walled components in boiler / turbine	Integrate thermal mitigation strategy.	Boiler fatigue monitoring system and Installation of EOH.	To be explored.
9	Milling Performance; No provision to measure fuel flow imbalance in mills	Mill performance mapping of coal flow in coal pipe at various load with	Milling system coal pipe measurement system installation for online balancing and	Coal pipe measurement available

		offline dirty pitot tube and balancing it.	combustion optimization.	Online balancing provision Currently not available, to be explored
10	Poor flame stability and load responsiveness with 40% - 55% high ash domestic Indian coal	Flame Quality Scanner Perf improvement; Optimizing mill operation with optimal Pri to Secondary air ratio.	Flame scanner upgrade in control, repositioning of scanners	In pipeline, Material awaited.
11	Chemical parameters control at start-up and load less than 55%	Improved strategy for start-up and part load operations	CPU installation	CPU available
18	REVIEW			
	Head of Operation shall review this document on a 2-yearly basis or as necessary			
19	DISTRIBUTION LIST			
	The approved soft copy will be maintained in EEMG website as controlled copy			

ANNEXURE

TEST PROCEDURES FOR A 500 MW UNIT, FOR RAMP DOWN FROM FULL LOAD AND VICE VERSA

A. Test procedure for load ramp down tests from full load to 300 MW (60%) load at different ramp rates in CMC - 1% per minute and 3% per minute:

1. Stabilize the unit load at full load with 6 mills.
2. Give the load set point of 450 MW.
3. Start reducing the mill loading of bottom most mill gradually to meet firing demand.
4. Stabilize the unit around 450 MW.
5. Give the load set point of 400 MW.
6. Start reducing the mill loading of bottom most mill gradually till the mill is completely unloaded. Trip the bottom most mill.
7. Stabilize the unit around 400 MW.
8. Give the load set point of 350 MW.
9. Start reducing the mill loading of bottom most mill gradually to meet firing demand.
10. Stabilize the unit around 350 MW.
11. Give the load set point of 300 MW.
12. Start reducing the mill loading of bottom most mill gradually till the mill is completely unloaded. Trip the bottom most mill.
13. Stabilize the unit around 300 MW.

B. Test procedure for load ramp down tests from 60% to 40% load at 1% & 3% per minute.

1. Stabilize the unit load at 60% load with 4 mills.
2. Give the load set point of 250 MW.
3. Start reducing the mill loading of all mills gradually with higher unloading of bottom most mill to meet firing demand.
4. Manually switchover to single BFP operation as further load ramp down is taken up.
5. Maintain both PA Fans in service, if fans are operating near to their stall zone then manually switchover to single PA Fan operation as further load ramp down is taken up.
6. Stabilize the unit around 250 MW.
7. Give the load set point of 200 MW.
8. Start reducing the mill loading of all mills gradually with higher unloading of bottom most mill to meet firing demand.
9. Stabilize the unit around 200 MW.

C. Test procedure for load ramp up tests from 40% load to 60% load at 1% & 3% per minute:

1. Stabilize the unit load at 40% load with 4 mills and lesser loading in bottom most mill.
2. Give the load set point of 250 MW.
3. Start increasing the mill loading of all mills gradually with higher loading of bottom most mill to meet firing demand.
4. Stabilize the unit around 250 MW.
5. Give the load set point of 300 MW.
6. Start increasing the mill loading of all mills gradually with higher loading of bottom most mill to meet firing demand.
7. Manually take second BFP also in service, if not taken earlier, and balance both BFP.
8. Manually take second PA Fan in service, if not taken earlier, and balance both fans.
9. Equalize loading of all mills.
10. Stabilize the unit around 300 MW

Test procedure for load ramp up tests from 60% load to full load at 1% & 3% per minute

1. Stabilize the unit load at 60% load with 4 mills.
2. Take the fifth mill in service (preferably adjacent to mills already in service and topmost amongst the standby mills) with minimum loading and allow the mill to stabilize.
3. Give the load set point of 350 MW.
4. Increase the mill loading of fifth mill gradually to meet firing demand. Simultaneously adjust the mill loading of other four mills to meet firing demand and till the time fifth mill is sufficiently loaded and stabilized.
5. Stabilize the unit around 350 MW.
6. Give the load set point of 400 MW.
7. Increase the mill loading of fifth mill gradually to meet firing demand. Simultaneously adjust the mill loading of other four mills to meet firing demand and till the time fifth mill is sufficiently loaded and stabilized.
8. Stabilize the unit around 400 MW and equalize the mill loading.
9. Take the sixth mill in service (preferably adjacent to mills already in service and topmost amongst the standby mills) with minimum loading and allow the mill to stabilize.
10. Give the load set point of 450 MW.
11. Increase the mill loading of sixth mill gradually to meet firing demand. Simultaneously adjust the mill loading of other five mills to meet firing demand and till the time sixth mill is sufficiently loaded and stabilized.
12. Stabilize the unit around 450 MW.
13. Give the load set point of 500 MW.
14. Increase the mill load of sixth mill gradually to meet firing demand. Simultaneously adjust the mill loading of other five mills to meet firing demand and till the time sixth mill is sufficiently loaded and stabilized. Stabilize the unit at full load and equalize the mill loading